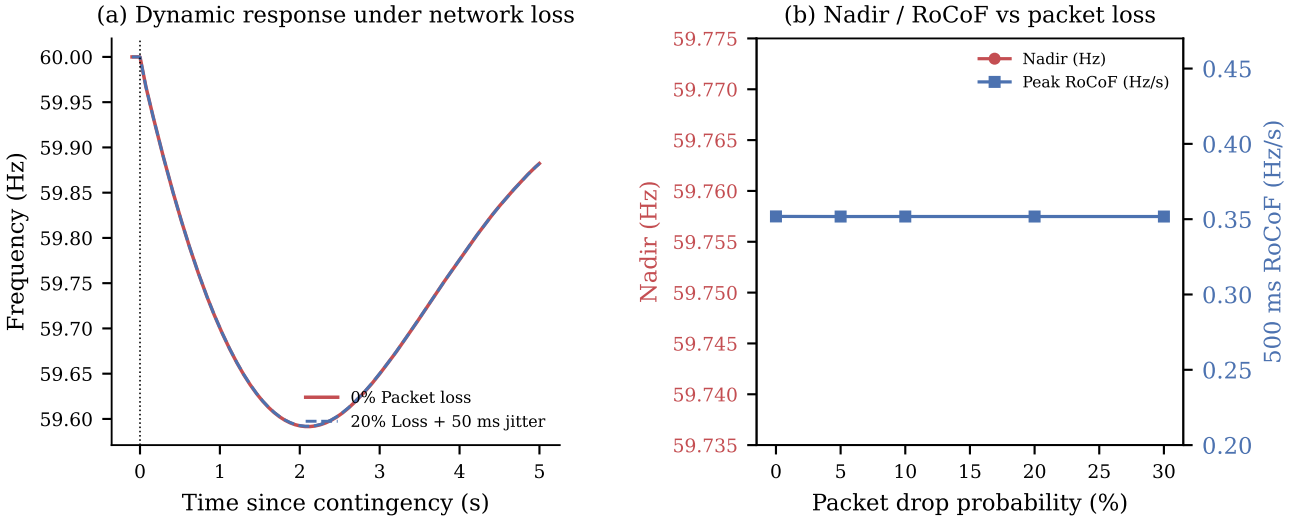
(a) Dynamic response under network loss
Frequency (Hz)
60.00
59.95
59.90
59.85
59.80
59.75
59.70
59.65
59.60
0% Packet loss
20% Loss + 50 ms jitter
0
1
2
3
4
5
Time since contingency (s)
(b) Nadir / RoCoF vs packet loss
Nadir (Hz)
Peak RoCoF (Hz/s)
Nadir (Hz)
59.775
59.770
59.765
59.760
59.755
59.750
59.745
59.740
59.735
500 ms RoCoF (Hz/s)
0.45
0.40
0.35
0.30
0.25
0.20
0
5
10
15
20
25
30
Packet drop probability (%)